Proprietary to General Electric Company
Direction 5438996-8EN, **Rev#** 9
Brivo CT385 Series Service Methods (Adv)
Module Version 3.0, **Version Date** 7/3/13

Tube Replacement

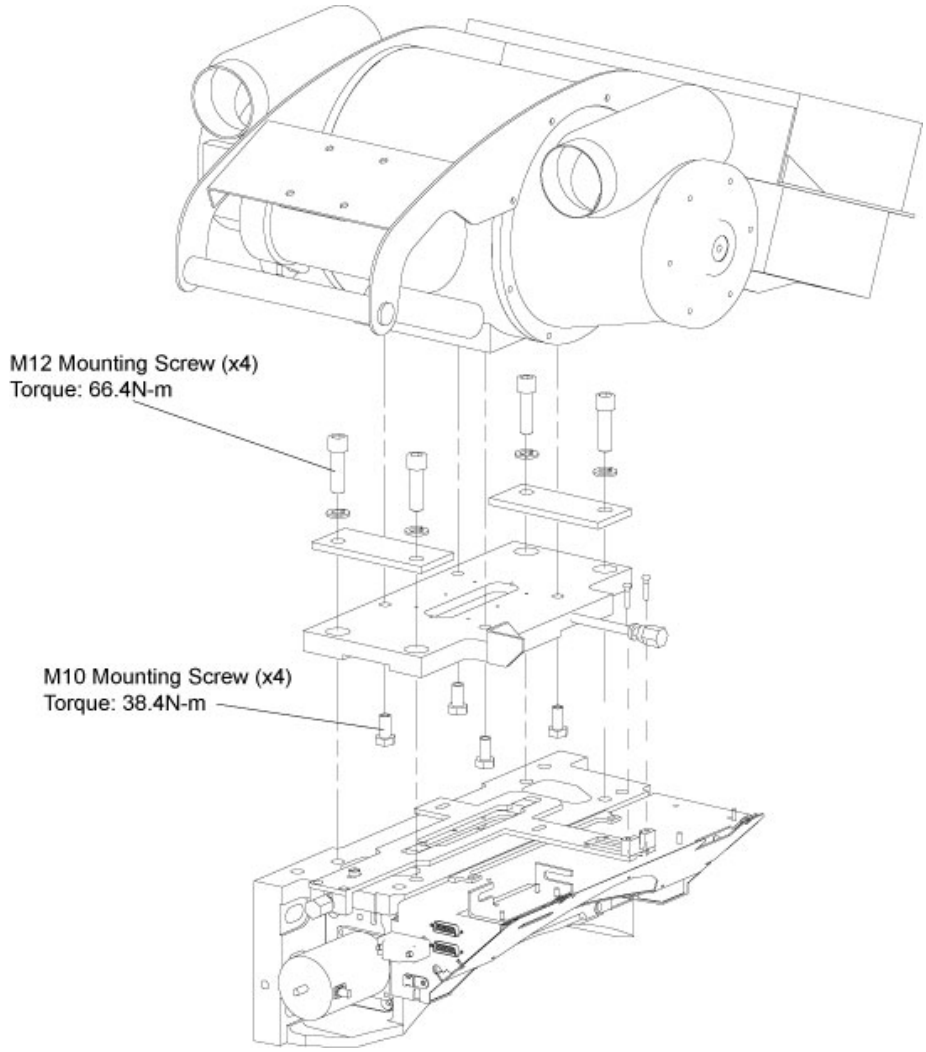
(Click Here for Summary Version)

1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	1 hour	7 hours

2 Overview

Illustration 1: Tube Removal Diagram



3 Preliminary Requirements

3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
HV Bleeder Kit	1	-	2345332	-
Hoist - 2 - PC, ARM	1	-	2282509-2	-
Hoist - 2 - PC, Long Tube	1	-	2282510-2	-
Chain Block (Hoist Assembly)	1	-	2268140	-
Alignment JIG Holder	1	-	5371128	-
Alignment JIG	1	-	5371129	-
Dial Gauge Indicator (millimeters or inches)	2	-	-	-
Spanner Wrench	1	-	-	-

3.2 Consumables

Item	Qty	Effectivity	Part#	Manufacturer
Tube Oil	AR	-	-	-
Tie-wrap	AR	-	-	-
Dry Wipes or Paper Towel		-	-	-

3.3 Replacement Parts

Item	Qty	Effectivity	Part#	Manufacturer
Tube	1	-	5363417	-

3.4 Safety



⚠ WARNING

POTENTIAL FOR ELECTRIC SHOCK.
LETHAL VOLTAGES MAY BE PRESENT IN THE GANTRY.
FOLLOW PROPER ELECTRICAL LOCK-OUT/TAG-OUT PROCEDURES.



⚠ WARNING

POTENTIAL FOR PERSONAL INJURY (PINCH, CRUSH, ENTANGLEMENT OR IMPACT INJURIES ARE POSSIBLE).
GANTRY HAS ROTATING ASSEMBLY.
USE THE GANTRY'S ROTATIONAL LOCK, TO PREVENT UNCOMMANDED MOTION OF THE GANTRY'S ROTATING ASSEMBLY.



⚠ CAUTION

BURN/SHOCK HAZARD.
Wait more than fifteen minutes following the last scan before performing the following procedure.



NOTICE

Potential HV Cable Damage.

Do not sharply bend HV cables, or damage the cables. The minimum allowed radius of bent HV cables is 6.35 cm (2.5 in.).



NOTICE

Follow ALL required safety and PPE procedures customary for your organization, when working on this product.

3.5 Required Conditions

Condition	Reference	Effectivity
Only trained service personnel should service the GE CT Scanner.	-	-

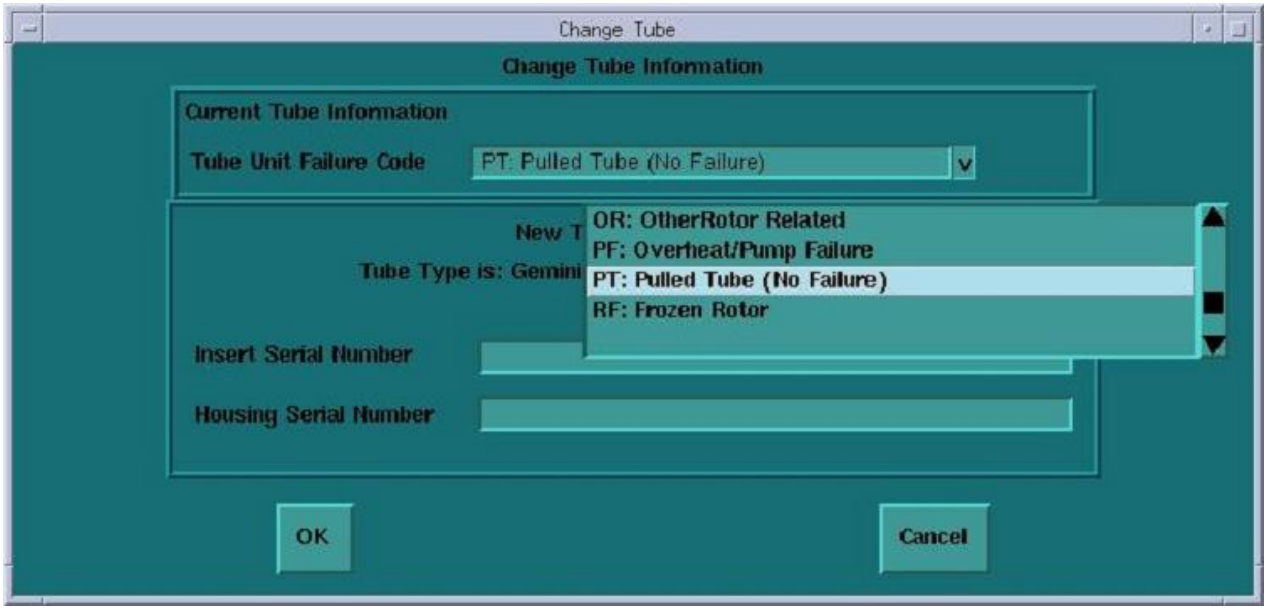
4 Procedure

4.1 Data Entry

NOTE: The following procedure is required prior to the actual tube replacement. If you perform the data entry after tube replacement, the system will set the tube heat data to the maximum, and you will have to wait for a pretty long time until the tube heat data lowers.

1. Open [SERVICE DESKTOP].
2. Enter the data of the new tube:

   Select [REPLACEMENT] → [NEW TUBE CONFIG], select a proper Tube Unit Failure Code (reason for tube replacement), Insert Serial #, and Housing Serial # of a new tube, then click on OK.



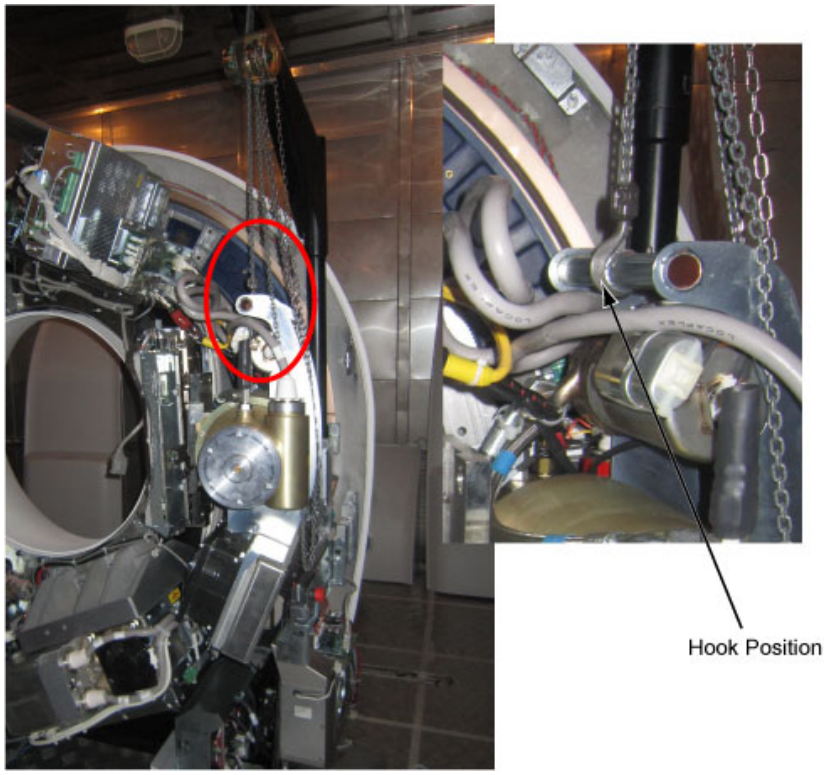
4.2 Old Tube Removal

⚠ DANGER

ELECTROCUTION HAZARD.
HIGH VOLTAGES PRESENT.
USE PROPER LOCKOUT/TAGOUT PROCEDURES BEFORE REPLACING COMPONENTS. SEE SAFETY MENU OF SERVICE DOCUMENT GUI FOR APPROPRIATE PROCEDURES.

1. Position the table to its lowest position.
2. Remove the gantry left side cover.
3. Turn off the AXIAL DRIVE, HVDC and 120VAC switches at the service switch panel.
4. Remove the table foot switch cover.
5. Remove the gantry side, top and front covers.
6. Turn on the 120VAC switch and rotate the gantry until the tube reaches about 3 o'clock position.
7. Insert the rotational lock pin, then turn off the 120VAC switch.
8. Insert the long tube and hoist arm, then attach the hoist chain to the x-ray tube and pull in all loose chain until the chain is taut.

Illustration 2: Hoist Assembly Installation



NOTE: The long tube must be fully inserted into the bracket, otherwise it may cause injury where least expected.

Correct



Incorrect

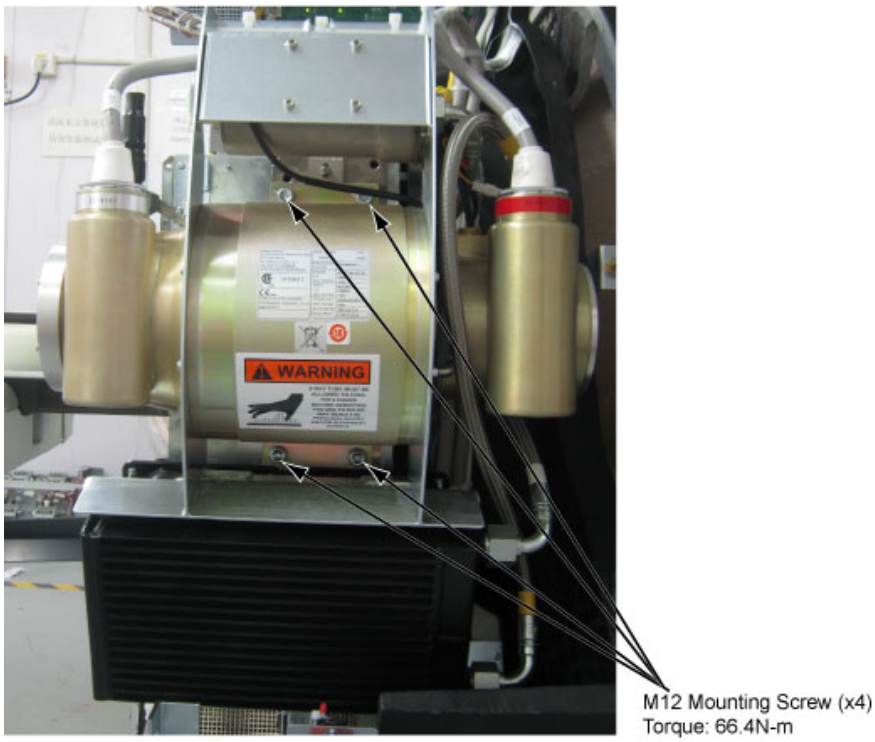


9. Disconnect the yellow pump power cable (5179363) and the rotor control cable (5432032) connectors from the tube.
10. Carefully cut tie-wraps securing HV cables. Note HV cables routing.
11. Use the spanner wrench to loosen the HV cable locking ring and remove the cable leads (anode/cathode) from the receptacle.

    NOTE: Take care not to damage the HV cable leads.
12. Ground the end of the cables to the gantry frame.
13. Wipe any residual oil from the leads to avoid dripping and insert kimwipe into the receptacle to prevent oil from leaking.
14. Unscrew the POR adjuster screw on the collimator.
15. Remove the tube assy from the gantry by unscrewing four M12 mounting screws. (See Illustration 1)

    Dispose these four mounting screws, do not use them for the new x-ray tube.

Illustration 3: X-Ray Tube Removal



16. Use the hoist assy to slowly lower the x-ray tube to the ground.
17. Remove the tube mounting plate (5403961) from the tube by unscrewing four M10 mounting screws. (See Illustration 1)

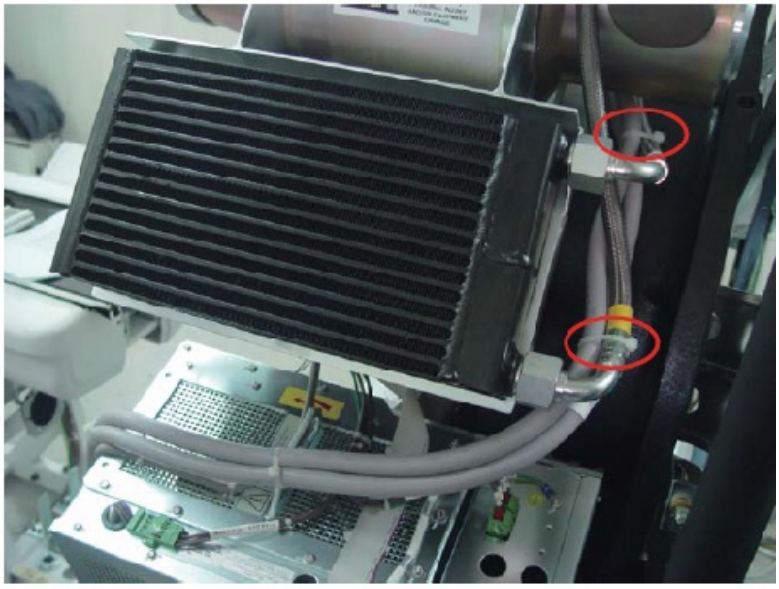
    Dispose these four plate screws, do not use them for the new x-ray tube.

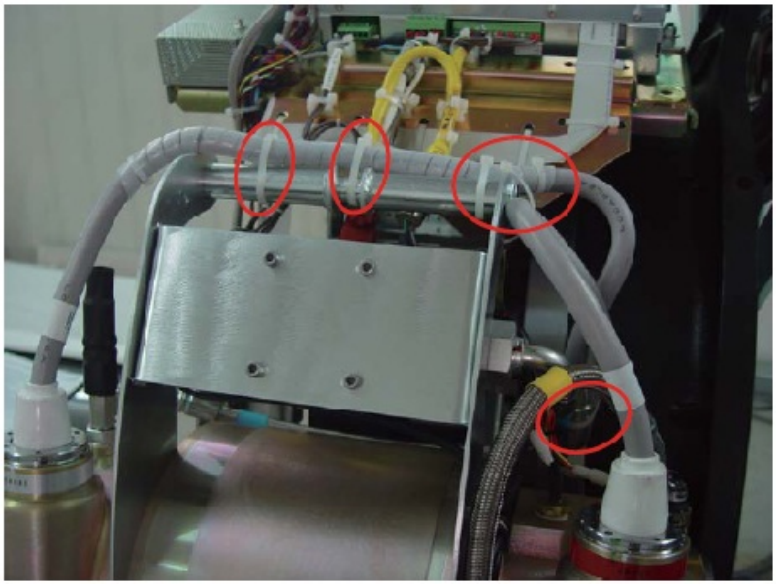
4.3 New Tube Installation

1. Install the tube mounting plate to the new x-ray tube with new screws (M10 x 1.5 x 20mm, torque: 38.4N-m). Then use the hoist to raise the new x-ray tube into place.
2. Mount the new x-ray tube by using new screws (M12 x 1.75 x 40mm) to the gantry. (Carefully with the x-ray tube secure: Torque: 66.4N.m)
3. Reinstall the HV cables (cathode/anode) as follows:
    a. Inject oil into the tube receptacles well to avoid air gaps when HV cable leads are inserted.
    b. Insert the HV cable leads into the tube receptacles and tighten the cable locking ring with the spanner wrench.
    c. Wipe excess oil from the tube.
4. Reconnect the pump power cable and the rotor control cable connectors to the tube.
5. Re-tie any tie-wraps which were removed in Old Tube Removal, Step 10.

NOTE: Incorrectly routed or secured HV cables will result in damage to the HV cables and/or other parts of the gantry.

Illustration 4: X-Ray Tube Tie-Wrap Configuration





6. Remove the hoist chain, hoist arm and long tube from the x-ray tube and gantry.
7. Check for oil leak, if oils leak is found, tighten the locking ring slowly until there is no leakage, paying attention to not over tighten.
8. Turn on the 120VAC switch, verify that the radiator fan for the x-ray tube cooling and JEDI power unit fan are operating normally.
9. Turn on the AXIAL DRIVE and HVDC switches at the service switch panel. Wait at least 10 minutes to warm up the filament.

5 Finalization

1. Check new mounting bolts and washers are used for mounting the tube; old mounting bolts and washers are discarded.

2. Perform the retest steps below in the order described.
    a. Install TubeInstallFlex option through eLicense, as needed. Refer to Tube Identification (Tube Alignment Workflow)
    b. *Gantry Rotation Safety Check*
    c. *Reset TnT Data (Generator Tool)*
    d. *Plane of Rotation (POR) Alignment Process*
    e. *Beam on Window (BOW) Alignment Process*
    f. *CBF/SAG Alignment Process*
    g. *ISO Alignment Procedure*
    h. *Meter Verification*
    i. *HV Tank Feedback Resistor Verification*
    j. *Filament Calibration*
    k. *HHS Scans (Adv)*

       Tube Usage Information is found on the Common Service Desktop following path:

       [Error Logs] → [Tube Usage] → [Details].
    l. *Gantry Balance Procedure*
    m. *Hot ISO Alignment*
    n. *Collimator Calibration*
    o. *Detailed Calibration*
    p. N Number Check
    q. *System Scanning Test*
    r. *Quality Assurance Test*
    s. *New Tube Configure*
    t. Save System State

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